

PROJECT 7: ENABLE MONITORING AND LOGS USING AMAZON CLOUDWATCH REPORT

INTRODUCTION

Amazon CloudWatch is a monitoring and observability service provided by Amazon Web Services (AWS). It enables users to collect, monitor, and analyze performance metrics, logs, and events from AWS resources and applications. CloudWatch helps administrators maintain the health, availability, and performance of cloud infrastructure by providing real-time monitoring and automated alerts.

This project focuses on monitoring an Amazon EC2 instance using Amazon CloudWatch. The implementation includes launching an EC2 instance, viewing CloudWatch performance metrics, creating a CloudWatch alarm based on CPU utilization, and verifying the alarm status. The project demonstrates how CloudWatch can be used to proactively monitor cloud resources and improve operational reliability.

OBJECTIVES

The primary objectives of this project are:

- ✧ Launch an Amazon EC2 instance
- ✧ Monitor EC2 performance metrics using Amazon CloudWatch
- ✧ View CPU utilization and network metrics
- ✧ Create a CloudWatch alarm for CPU utilization
- ✧ Verify the alarm status
- ✧ Understand the fundamentals of AWS monitoring and alerting
- ✧ Learn the importance of proactive infrastructure monitoring

OVERVIEW OF AMAZON CLOUDWATCH

Amazon CloudWatch

Amazon CloudWatch is a monitoring and management service that provides data and actionable insights for AWS resources, applications, and services. It collects performance metrics, logs, and events, allowing administrators to monitor system health and respond to operational issues.

CloudWatch Metrics

CloudWatch Metrics are time-ordered data points collected from AWS services. For Amazon EC2, CloudWatch automatically provides metrics such as CPU Utilization, Network In, Network Out, Disk Read Bytes, and Disk Write Bytes.

CloudWatch Alarms

CloudWatch Alarms continuously monitor selected metrics and compare them against predefined thresholds. When the threshold is exceeded, the alarm changes its state and can trigger notifications or automated actions.

CPU Utilization

CPU Utilization measures the percentage of allocated compute resources currently being used by an EC2 instance. Monitoring CPU utilization helps identify overloaded or underutilized instances and supports efficient resource management.

METHODOLOGY

Step 1: Launching the EC2 Instance

An EC2 instance named **CloudWatch-Monitoring-Server** was launched using the AWS Management Console. The Amazon Linux 2023 Amazon Machine Image (AMI) was selected along with the t3.micro instance type. A key pair and security group were configured, and the instance was successfully launched and verified to be in the running state.

Step 2: Monitoring EC2 Metrics

The Monitoring tab of the EC2 instance was opened to observe CloudWatch metrics. Performance metrics including CPU Utilization, Network In, Network Out, and Network Packets were viewed. After a few minutes, CloudWatch successfully collected and displayed the monitoring data.

Step 3: Creating a CloudWatch Alarm

Amazon CloudWatch was opened from the AWS Management Console. A CloudWatch alarm was created using the **CPUUtilization** metric for the EC2 instance. The alarm was configured with the following settings:

- ✧ Metric: CPUUtilization
- ✧ Threshold Type: Static
- ✧ Condition: Greater than 80%
- ✧ Evaluation Period: 5 minutes
- ✧ Statistic: Average

The alarm was successfully created and associated with the EC2 instance.

Step 4: Verifying the Alarm

The created alarm was viewed in the CloudWatch Alarms dashboard. Initially, the alarm state was displayed as **Insufficient Data** while CloudWatch collected monitoring information. After receiving sufficient metrics, the alarm state automatically changed to **OK**, confirming successful monitoring and alarm configuration.

Step 5: Resource Cleanup

After completing the project, the EC2 instance was terminated to avoid unnecessary AWS usage. The CloudWatch alarm was also deleted, ensuring that no unused monitoring resources remained in the AWS account.

RESULTS

The following outcomes were successfully achieved:

- ✧ Launched an Amazon EC2 instance
- ✧ Successfully viewed CloudWatch monitoring metrics
- ✧ Monitored CPU Utilization and network metrics
- ✧ Created a CloudWatch alarm based on CPU utilization
- ✧ Verified the alarm status as **OK**
- ✧ Gained practical experience in AWS monitoring and alert configuration
- ✧ Cleaned up AWS resources after project completion

KEY LEARNINGS

Throughout this project, several important cloud monitoring concepts were learned:

- ✧ Amazon CloudWatch provides real-time monitoring for AWS resources.
- ✧ CloudWatch Metrics help analyze the performance of EC2 instances.
- ✧ CloudWatch Alarms automatically monitor metrics against configured thresholds.
- ✧ CPU Utilization is an important performance metric for EC2 instances.
- ✧ Monitoring improves system reliability and operational visibility.
- ✧ AWS provides automated monitoring without additional infrastructure.
- ✧ Cleaning up cloud resources helps prevent unnecessary costs.

ADVANTAGES

Amazon CloudWatch provides several benefits:

- ✧ Real-time infrastructure monitoring
- ✧ Automated performance tracking
- ✧ Easy alarm configuration
- ✧ Improved system reliability
- ✧ Seamless integration with AWS services
- ✧ Proactive issue detection
- ✧ Scalable monitoring for cloud resources

LIMITATIONS

Despite its advantages, CloudWatch also has certain limitations:

- ✧ Basic monitoring provides limited metrics compared to detailed monitoring.
- ✧ Alarm effectiveness depends on correctly configured thresholds.

- ✧ Additional charges may apply for advanced monitoring features.
- ✧ Continuous monitoring alone cannot resolve infrastructure issues.
- ✧ Incorrect alarm settings may produce unnecessary alerts.
- ✧ Some advanced monitoring requires CloudWatch Agent installation.

CONCLUSION

This project provided practical hands-on experience in monitoring AWS resources using Amazon CloudWatch. The implementation involved launching an EC2 instance, viewing CloudWatch metrics, configuring a CPU utilization alarm, verifying its status, and cleaning up cloud resources after completion.

The successful completion of this project demonstrated how Amazon CloudWatch can be used to monitor cloud infrastructure and provide automated alerting based on resource performance. It also reinforced important cloud computing concepts such as infrastructure monitoring, performance analysis, alarm configuration, and operational management.

Overall, the project strengthened practical AWS monitoring skills and provided valuable experience in implementing cloud observability and proactive resource management.